Deep Patel

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PROFESSIONAL SUMMARY

High school senior focused on aerospace engineering and rocket propulsion. Creator of ORBIX, an aerospace engineering web app, and designer of a 3D-printed rocket bell nozzle studied with CFD simulation. Dedicated youth leader and volunteer looking for opportunities to learn more about physics, engineering, and space science through hands-on projects and research.

EDUCATION

Chelmsford High School, Chelmsford, MA Expected Graduation: June, 2027

GPA: 4.01 (Weighted)

Relevant Courses: AP Physics C, AP Computer Science, Calculus, Advanced Engineering Principles & 3D Design

SKILLS AND INTERESTS

Technical Skills: Java, CAD (Fusion 360, Revit, Onshape, OpenSCAD), CFD Simulation (SimFlow), Robotics, Microsoft Office Suite, Mathematical Modeling, Adobe Creative Cloud

Interests: Aerospace Engineering, Mechanical Engineering, Astrophysics, Rocketry, 3D Modeling

TECHNICAL PROJECTS

Aerodynamic Bell Nozzle Sep. 2025 - Present

Lead Designer & Fabricator (Self-Directed)

- Studied nozzle thermodynamics and isentropic flow independently to size a LOX/RP-1 bell nozzle for a 100 bar chamber expanding to 0.7 bar.
- Modeled the nozzle contour in OpenSCAD and checked its 15.92 area ratio against the Mach 3.6 isentropic design point.
- Set up and debugged a compressible-flow CFD case in SimFlow, tracing solver divergence through mesh, boundary condition, and limiter changes.
- 3D-printed the 84 mm nozzle in PLA, printing it inverted so no support material formed inside the bell.

ORBIX Web Application Aug. 2026 - Present

Creator & Project Lead | https://orbix-aerospace.vercel.app/

- Conceived and designed a web app with 13 engineering calculators, a mission planner, and profiles of 10 real aircraft and launch vehicles.
- Wrote the specifications for each tool, including equations, inputs, and units, and directed the programming of the front-end logic and physics engines using AI coding tools.
- Planned a verification page that checks results against published references (NACA Report 1135, U.S. Standard Atmosphere 1976); 26 of 36 values match within rounding.

ACADEMICS AND EXTRACURRICULARS

Track and Field Team Nov. 2023 - Present

Captain & Athlete

- 3x MSTCA Div. II State Relay Medalist (High Jump State Relay: 4th, 4th, 6th) and 6-season varsity athlete (Indoor & Outdoor).
- Compete in high jump, 55m hurdles, and 110m hurdles. Lead daily workouts for 75+ athletes and mentor 35+ underclassmen on form and discipline.

Artificial Intelligence (AI) Club Sep. 2025 - Present

Vice President

- Manage 15 members and host bi-weekly meetings focused on safe AI usage and coding ethics in the real world.
- Directed a project team for the MIT URTC conference, building a custom AI software tool that monitors home environments to optimize care for household plants.

Robotics Club Sep. 2023 - Jun. 2024

Member

- Worked with a team to design, build, and test robots for VEX Robotics competitions.
- Programmed the robot to navigate competition mazes autonomously and engineered the physical mechanisms needed to lift plastic balls into scoring baskets.

WORK EXPERIENCE

ALPHAPEP LLC, Lowell, MA May 2026 - Present

Email Marketing Associate

- Design and distribute targeted marketing emails to advertise a diverse variety of the company's products.
- Manage and execute outgoing email campaigns to ensure successful delivery of promotional materials and drive customer engagement.

LEADERSHIP AND VOLUNTEERING

Bachat Kumbh Project, Lowell, MA — Sep. 2025 - Present

Co-Founder & Logistics Lead

- Bridged engineering and community service by designing and 3D-printing custom, temple-shaped donation boxes (Kumbhs) for local youth.
- Managed the distribution, tracking, and collection logistics of the physical boxes across the community.
- Successfully mobilized youth participation to raise $2,500+ to directly fund the construction of the new Tyngsborough temple.

BAPS Swaminarayan Sanstha (Bal Mandal & Kishore Mandal), Lowell, MA — Aug. 2022 - Present

Youth Activities Coordinator & Volunteer — *Aug. 2024 - Present*

Teacher — *Aug. 2022 - Aug. 2024*

- Lead weekly youth assemblies for 70+ Pre-K–8th grade children and manage a team of 15 high school peer teachers.
- Plan and oversee weekly lessons covering cultural values, history, and leadership skills.
- Serve as a logistics volunteer for the high school teen wing (Kishore Mandal), helping coordinate stage setup and crowd control for major cultural festivals serving over 1,000 attendees.

BAPS Charities, Lowell, MA — Jun. 2017 - Present

Walkathon Organizer & Outreach Volunteer

- Organized the annual 'In the Joy of Others' Walk-Run for 650+ participants, setting up routes and managing event registration.
- Helped raise more than $165,000 over four years for the Lowell Police Dept., Tyngsborough Fire Dept., local public schools, and The Nature Conservancy's Plant a Billion Trees campaign through the Walk Green initiative.

BAPS Akshardham, Robbinsville, NJ — Jul. 2023 - Oct. 2023

Full-Time Volunteer

- Designed detailed architectural pieces using CorelDRAW and Adobe Illustrator, and created CNC machine files for a large cultural building project.
- Ran industrial CNC machines and hot wire cutters to make complex parts out of wood, PVC, and foam at scale.
- Worked with senior designers and other volunteers to ensure designs were accurate and finished on time without wasting materials.